

**Name: Y.Sreenadh**

**Reg no: 192525166**

## 29. Classical Process Synchronization Problem

### Code:

```
#include <stdio.h>

int mutex = 1, full = 0, empty = 5; int
item = 0;

void producer()
{   if(mutex == 1 && empty >
0)
    {        mutex
= 0;
item++;
full++;
empty--;

        printf("Producer produces item %d\n", item);

        mutex = 1;
    }   else        printf("Buffer
is full.\n");
}

void consumer()
{
    if(mutex == 1 && full > 0)
    {        mutex = 0;        printf("Consumer
consumes item %d\n", item);
        item--
;        full--
;
        empty++;

        mutex = 1;
    }
    else
    printf("

```

```

Buffer
is
empty.\
n");
}

int main()
{
    int
choice;

    printf("1. Producer\n2. Consumer\n");
    printf("Enter choice: ");    scanf("%d",
&choice);

    if(choice == 1)
producer();    else if(choice ==
2)    consumer();    else
printf("Invalid choice.\n");

    return 0;
}

```

**Output:**

```

1. Producer
2. Consumer
Enter choice: 1
Producer produces item 1

...Program finished with exit code 0
Press ENTER to exit console.

```